PAPER ID-310218

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2024-25
SOLID WASTE MANAGEMENT

TIME: 3 HRS **M.MARKS: 100**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	CO	Level
a.	What is integrated solid waste management?	1	K2
b.	What are different types of waste generated from a rural household, write in increasing order of quantity?	1	K2
c.	What are different transportation equipment used for carrying solid waste in urban areas?	2	K2
d.	What do you mean by processing of solid waste?	2	K2
e.	Define the term "Land filling".	3	K2
f.	Draw the flowchart of landfill classification.	3	K2
g.	Distinguish between incineration and pyrolysis.	4	K2
h.	Define composting.	4	K2
i.	Explain corrosivity.	5	K2
j.	Differentiate between biomedical and hazardous waste.	5	K2

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 20**

Q no.	Question	CO	Level
a.	What is solid waste management process? Explain different levels in detail.	1	K5
b.	How solid waste is a global menace? What is the role of transportation in the mitigation of this problem? Explain using suitable example.	2	K4
c.	What are the challenges and opportunities associated with waste management in India?	3	K5
d.	Explain with the help of flow diagram, energy recovery system from solid waste.	4	K4
e.	Examine what are the harmful effects/risks involved due to hazardous waste.	5	K4

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

Q no.	Question	CO	Level
a.	What is the difference between hauled and stationary container system? Explain using neat sketches.	1	K3
b.	Design a traditional hauled container system using the data given below: K3, K4, K5 Time spent to travel between two container location = 25 minutes/trip. Time spent at one container location for loading and unloading of filled	1	K5

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	and emptied container respectively=0.6 h/trip, Empirical haul time constant= 0.022 h/trip, and the average one way distance between container and processing site is 18 miles/trip with speed limit of 45 miles/h, Time spent at site=0.133h/trip. Determine the number of containers that can be emptied in a 9 hour work period.		

4. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Write a short note on engineering system for on-site handling and processing of solid waste.	2	K3
b.	Explain following processing technique: 1) Magnetic separation 2)Air Separation	2	K2

5. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	What are the different phases in life cycle of land filling?	3	K2
b.	Discuss various factors affecting the site selection of land filling? Give the advantage and disadvantage of land filling.	3	K3

6. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Discuss the composting and their types.	4	K2
b.	Discuss aerobic and anaerobic decomposition of waste.	4	K2

7. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Name the six characteristics of the waste because of which any waste can be classified as hazardous waste (regardless of their concentration limits).	5	K3
b.	Discuss the various techniques used for disposal of hazardous waste.	5	K2

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